

Notes: Carpenter, Lu, and Whitelaw (JFE, 2021)

The Real Value of China's Stock Market

Contents

1 Big picture

2 Data and measurement (key objects)

- 2.1 Why this setting is useful
- 2.2 Data sources and sample construction
- 2.3 How the main objects are measured
- 2.4 Three model-based outcome measures
- 2.5 Institutional facts that matter for interpretation

3 Models and empirical specifications (all equations + interpretation)

- 3.1 Price informativeness about future profit
- 3.2 Price informativeness about future investment
- 3.3 Allocational efficiency
- 3.4 State ownership and future-profit informativeness
- 3.5 Dual listing and QFII ownership
- 3.6 State ownership and price informativeness about future investment
- 3.7 State ownership and allocational efficiency
- 3.8 Return-based cost-of-capital evidence

4 Identification and what makes the estimates credible

- 4.1 What identifies the paper's main facts
- 4.2 Why the results are not obviously mechanical
- 4.3 Why the institutional story is plausible
- 4.4 What the paper cannot fully establish

5 Tables and figures

- 5.1 Figure 2: China's prices become about as informative as US prices
- 5.2 Figure 4: the rise in informativeness lines up with reform
- 5.3 Table 2: state ownership becomes a major drag after 2008
- 5.4 Figure 5: private firms use the market more than SOEs
- 5.5 Tables 8 and 9: China's market offers high returns and low correlation, which implies a high cost of capital

6 Short summary

7 Conclusion

7.1 Core conclusion

7.2 Economic intuition

7.3 Takeaway

1 Big picture

- **Question.** What capital-allocation role can China's stock market play? Does it aggregate useful information about firms' future profitability, does that information feed into investment decisions, and what does the market imply about firms' cost of equity capital?
- **Setting.** The paper studies China's A-share market from 1995 to 2016 and compares China's price informativeness to the US. Within China, it also contrasts privately owned firms with state-owned enterprises (SOEs).
- **Why this matters.** China's economy is investment driven and its financial system is dominated by state-owned banks. If the stock market can transmit informative price signals, it can become a second capital-allocation channel rather than a speculative side market.
- **Conventional wisdom the paper challenges.** China's stock market is often described as a "casino" dominated by retail trading, regulation, and noise. The paper argues that this characterization became much less accurate after the early-2000s reform wave.
- **Core conceptual contribution.** The paper brings the Bai et al. (2016) framework to China. It separates *forecasting price efficiency* (FPE: do prices forecast future profits?) from *revealtory price efficiency* (RPE: do prices reveal information that managers use to improve real decisions?).
- **Main empirical strategy.** Estimate yearly cross-sectional regressions linking future profits to current valuation ratios, future investment to current valuation ratios, and future profits to current investment. Then study how those links vary with state ownership, dual listing, and foreign ownership.
- **Main result on price informativeness.** Stock prices in China were weakly informative in the late-1990s/early-2000s "casino" period, but since about 2004 they have been about as informative about future profits as stock prices in the US.
- **Main real-economy result.** For privately owned firms, the rise in price informativeness coincides with stronger investment-price sensitivity and better allocational efficiency, consistent with managers learning from market prices.
- **SOE result.** After the 2008–2009 stimulus, price informativeness and investment efficiency for firms with high state ownership fall below those of private firms, consistent with subsidies and policy-directed investment making SOE profits harder to predict and less tied to value-maximizing choices.
- **Market-integration result.** Dual listing in Hong Kong is associated with *lower* A-share price informativeness, while QFII ownership is weakly positively related to informativeness.
- **Cost-of-capital result.** Realized market returns suggest an equity premium in China roughly 5 percentage points above the US and a monthly alpha of almost 1% in USD terms relative to US and global factors. The paper treats this evidence as suggestive rather than conclusive because realized returns need not equal expected returns.
- **Economic takeaway.** China's stock market appears capable of aggregating useful information and improving capital allocation for private firms. The main constraints are not an inherent inability of prices to convey information, but state intervention and incomplete market integration.

2 Data and measurement (key objects)

2.1 Why this setting is useful

- China is the world’s second-largest stock market, but the paper studies it precisely because its economic role is contested. The key question is not whether the market is large, but whether it is *useful*.
- China’s institutional environment provides strong variation in how informative prices should be: WTO-era reforms, the 2005 Split-Share Structure Reform, the introduction of QFII, margin trading, and short-selling pilots all plausibly affect the information content of prices.
- The coexistence of private firms and SOEs is especially helpful. The same market generates price signals for firms that plausibly maximize value and firms that may also pursue employment, subsidy, or policy goals.

Interpretation. This is not just a paper about whether Chinese prices forecast earnings. It is a paper about whether price signals matter more when firms are organized to use them.

2.2 Data sources and sample construction

- **Firm-level market and accounting data:** China Stock Market and Accounting Research (CSMAR), 1995–2016.
- **State ownership and QFII ownership:** CSMAR plus Wind. Wind is particularly important after the 2005 Split-Share Structure Reform because it tracks top state holders after many formerly nontradable shares become tradable.
- **US comparison:** the paper computes the same Bai et al. (2016)-style informativeness measures for US firms to benchmark China’s market.
- **Sample restrictions:** financial firms are excluded. Nominal variables are deflated by the GDP deflator. Variables are winsorized at the 1st and 99th percentiles. Industry controls use a one-digit SIC-style classification built from CSMAR industrial code B.
- **Inference:** the paper reports White heteroskedasticity-consistent cross-sectional t-statistics; industry-clustered standard errors are similar.

Interpretation. The design is transparent and heavily panel-based, but not quasi-experimental. The credibility of the results therefore comes from internal consistency, benchmarking to the US, and a sequence of robustness checks rather than from a clean external shock.

2.3 How the main objects are measured

- **Profit, $E_{i,t}$.** Net profit earned over calendar year t .
- **Market value, $M_{i,t}$.** End-of-year A-share price times total shares outstanding, including tradable A, B, and H shares plus nontradable shares.
- **Assets, $A_{i,t}$.** Book assets, used as the scaling variable throughout.

- **Investment, $I_{i,t}$.** Capital expenditure.
- **Valuation ratio.** $\log(M/A)$, which captures how strongly market values differentiate firms with different future prospects.
- **State ownership, $X_{i,t}$.** Pre-2005, the fraction of nontradable state-owned shares from CSMAR. Post-2005, the number of shares held by top-10 state entities from Wind, including both tradable and nontradable holdings.
- **Private firm versus SOE split.** Firms with state ownership of 40% or less are treated as private; firms above 40% are treated as SOEs.
- **Dual listing.** Indicator that an A-share firm also has a twin H-share listed in Hong Kong.
- **Foreign ownership.** Fraction of shares held by qualified foreign institutional investors (QFIIs).

Interpretation. The paper’s key variables are not just raw accounting outcomes. They are building blocks for model-based “predicted variation” measures intended to map market prices into managerial information and real efficiency.

2.4 Three model-based outcome measures

- **Price informativeness about future profit.** The predicted variation $b_t \times \sigma_t(\log(M/A))$ from a regression of future profit on current valuation and current profit. This is the paper’s measure of *forecasting price efficiency*.
- **Price informativeness about future investment.** The predicted variation $b_t \times \sigma_t(\log(M/A))$ from a regression of future investment on current valuation, profit, and investment. This tracks how strongly prices forecast future investment choices.
- **Efficiency of capital allocation.** The predicted variation $b_t \times \sigma_t(I/A)$ from a regression of future profit on current investment and current profit. If managers learn from prices and invest accordingly, this measure should move with price informativeness.

Interpretation. The paper does not measure “informativeness” by R^2 or idiosyncratic volatility. It measures it by how much current prices forecast economically relevant future outcomes.

2.5 Institutional facts that matter for interpretation

- Foreign investors still hold only about 3% of China’s A-share market, so the market remains overwhelmingly domestic.
- QFII ownership is tiny: the cross-sectional average never exceeds about 0.2%, and even the 75th percentile is zero in every year.
- Before the 2005 reform, state ownership is large and concentrated in nontradable shares. After the reform, mean state ownership falls below 20% and the median moves close to zero, producing a more right-skewed ownership distribution.
- Reverse mergers exist but do not seem large enough to drive the core results; excluding the smallest 10% or 30% of firms has little effect.

Interpretation. The paper’s results are best read as describing a market that becomes more informative as it broadens and modernizes, but whose signals are still distorted where the state remains a dominant owner.

3 Models and empirical specifications (all equations + interpretation)

3.1 Price informativeness about future profit

The core Bai et al. (2016) regression is

$$\frac{E_{i,t+k}}{A_{i,t}} = a_t + b_t \log\left(\frac{M_{i,t}}{A_{i,t}}\right) + c_t \frac{E_{i,t}}{A_{i,t}} + d_{ts} 1_{s_{i,t}} + \varepsilon_{i,t+k}.$$

The paper summarizes yearly informativeness by

$$\text{FPE}_t = b_t \times \sigma_t\left(\log\left(\frac{M}{A}\right)\right).$$

Interpretation. If market values do a good job distinguishing future winners from future losers, then the valuation ratio should forecast future profits. The measure is larger when there is both (i) more cross-sectional spread in valuations and (ii) a stronger mapping from valuations to future profits.

3.2 Price informativeness about future investment

The paper next estimates

$$\frac{I_{i,t+k}}{A_{i,t}} = a_t + b_t \log\left(\frac{M_{i,t}}{A_{i,t}}\right) + c_t \frac{E_{i,t}}{A_{i,t}} + d_t \frac{I_{i,t}}{A_{i,t}} + e_{ts} 1_{s_{i,t}} + \varepsilon_{i,t+k}.$$

The corresponding predicted-variation measure is again

$$\text{Price-to-investment signal}_t = b_t \times \sigma_t\left(\log\left(\frac{M}{A}\right)\right).$$

Interpretation. If managers are learning something useful from the stock market, firms with higher valuations today should invest more tomorrow, after controlling for current profitability and current investment.

3.3 Allocational efficiency

To test whether investment itself becomes more informative about future profits, the paper estimates

$$\frac{E_{i,t+k}}{A_{i,t}} = a_t + b_t \frac{I_{i,t}}{A_{i,t}} + c_t \frac{E_{i,t}}{A_{i,t}} + d_{ts} 1_{s_{i,t}} + \varepsilon_{i,t+k},$$

and defines

$$\text{Allocational efficiency}_t = b_t \times \sigma_t \left(\frac{I}{A} \right).$$

Interpretation. This is the paper's real-efficiency object. If firms that invest more today are precisely the firms that earn more tomorrow, then investment is being allocated toward higher-return opportunities.

3.4 State ownership and future-profit informativeness

The paper allows the valuation coefficient to vary with state ownership and with the pre/post-2009 split:

$$\frac{E_{i,t+k}}{A_{i,t}} = a_{0t} + a_{1t}X_{i,t} + (b_{0t} + (b_1 1_{t < 2009} + b_2 1_{t \geq 2009})X_{i,t}) \log \left(\frac{M_{i,t}}{A_{i,t}} \right) + c_t \frac{E_{i,t}}{A_{i,t}} + d_{ts} 1_{s_i,t} + \varepsilon_{i,t+k}.$$

It also estimates yearly cross-sectional versions with a year-specific interaction coefficient b_{1t} .

Interpretation. A negative b_2 means that, after 2008, price informativeness declines with state ownership. The timing matters because the paper links the divergence to the postcrisis stimulus and the heavier use of SOEs as policy vehicles.

3.5 Dual listing and QFII ownership

For each characteristic $X_{i,t}$ (dual-listing indicator or QFII ownership share), the paper estimates

$$\frac{E_{i,t+k}}{A_{i,t}} = a_{0t} + a_{1t}X_{i,t} + (b_{0t} + b_1 X_{i,t}) \log \left(\frac{M_{i,t}}{A_{i,t}} \right) + c_t \frac{E_{i,t}}{A_{i,t}} + d_{ts} 1_{s_i,t} + \varepsilon_{i,t+k}.$$

Interpretation. The dual-listing interaction asks whether links to Hong Kong improve or degrade A-share informativeness. The QFII interaction asks whether even small amounts of foreign institutional ownership are associated with more informative prices.

3.6 State ownership and price informativeness about future investment

The paper then estimates the state-ownership interaction version of the investment-price regression:

$$\frac{I_{i,t+k}}{A_{i,t}} = a_{0t} + a_{1t}X_{i,t} + (b_{0t} + (b_1 1_{t < 2009} + b_2 1_{t \geq 2009})X_{i,t}) \log \left(\frac{M_{i,t}}{A_{i,t}} \right) + c_t \frac{E_{i,t}}{A_{i,t}} + d_t \frac{I_{i,t}}{A_{i,t}} + e_{ts} 1_{s_i,t} + \varepsilon_{i,t+k}.$$

Interpretation. This tests whether SOEs become less responsive to price signals precisely when the state begins using them more aggressively for macroeconomic policy.

3.7 State ownership and allocational efficiency

Finally, the state-ownership interaction version of the investment-to-profit regression is

$$\frac{E_{i,t+k}}{A_{i,t}} = a_{0t} + a_{1t}X_{i,t} + (b_{0t} + (b_{1t}1_{t<2009} + b_{2t}1_{t\geq 2009})X_{i,t})\frac{I_{i,t}}{A_{i,t}} + c_t\frac{E_{i,t}}{A_{i,t}} + d_{ts}1_{s,i,t} + \varepsilon_{i,t+k}.$$

Interpretation. If b_2 is strongly negative, then the same unit of investment is much less informative about future profit among high-state-ownership firms after 2008. That is exactly what one would expect if investment is being directed for policy reasons rather than according to firm-specific profitability.

3.8 Return-based cost-of-capital evidence

The cost-of-capital section is intentionally simpler. The authors compute tradable-value-weighted monthly excess returns for China's market in CNY and USD, compare them to large developed markets, and then estimate USD alphas for China relative to US and global Fama–French–Carhart factor sets.

Interpretation. This is suggestive evidence, not a structural asset-pricing estimate. The key idea is that if China remains under-owned globally and poorly diversified domestically, firms may face a higher required return than comparable US firms.

4 Identification and what makes the estimates credible

4.1 What identifies the paper's main facts

- The paper does *not* rely on an instrument or discontinuity. Its evidence comes from annual cross-sectional forecasting relations and how those relations evolve over time and across ownership structures.
- The central identifying comparison is internal consistency: if prices become more informative, then investment should also become more sensitive to prices, and investment should become more informative about future profits.
- The private-firm versus SOE comparison sharpens interpretation because the Bai et al. logic is more appropriate when managers choose investment to maximize value.

Bottom line. The paper is strongest when it shows the same pattern in three places: prices forecast profits more strongly, prices forecast investment more strongly, and investment forecasts profits more strongly.

4.2 Why the results are not obviously mechanical

The authors directly confront several mechanical explanations.

- **Composition effects.** The sample expands sharply over time, so the paper checks whether higher informativeness is just coming from more dispersion in earnings or valuations. It finds no meaningful post-2003 composition effect of the kind Bai et al. document for the full US cross-section.
- **Shell value and reverse mergers.** Because Chinese listings may carry shell value, the authors exclude the smallest 10% and 30% of firms. The results barely move.
- **Special-treatment firms.** Excluding financially distressed “ST” firms also does not materially change the findings.
- **Accounting quality.** If Chinese accounting data are noisy, that should bias the paper *against* finding strong forecasting relations.

Interpretation. These checks do not make the design causal, but they do show the main findings are not fragile artifacts of sample composition or obvious market microstructure oddities.

4.3 Why the institutional story is plausible

- The increase in informativeness lines up with a sequence of reforms: WTO accession, stricter delisting rules, the National Nine Rules, QFII, the Split-Share Structure Reform, and the gradual introduction of margin trading and short selling.
- The SOE-private divergence begins after the 2008–2009 stimulus, exactly when the state channels financing through state-owned banks and leans more heavily on SOEs.
- Dual listing plausibly imports discount-rate shocks from Hong Kong into A-share valuations, while QFII ownership plausibly imports better information production.

Interpretation. The paper’s narrative is institutionally grounded. The timing of the results matches the timing of reforms and policy distortions in a way that is hard to ignore.

4.4 What the paper cannot fully establish

- **No sharp causal identification.** State ownership is not randomly assigned, so the SOE/private contrast is still partly descriptive.
- **Dual listing and QFII results are not causal.** More informative firms may choose foreign investors, and dual-listed firms differ systematically from other firms.
- **Realized returns are not expected returns.** The cost-of-capital evidence is explicitly preliminary because average realized returns over 1995–2016 may reflect unexpectedly strong performance.
- **The paper cannot fully distinguish two SOE channels.** Lower informativeness among SOEs may reflect unpredictable subsidies, direct investment mandates, or both.

Overall assessment. The paper is quite convincing on the descriptive claim that China’s market has become much more informative and that private firms appear to use those signals. It is more suggestive, but still persuasive, on the claim that policy distortions explain the post-2008 SOE divergence.

5 Tables and figures

5.1 Figure 2: China's prices become about as informative as US prices

Read:

- The series for China collapses around 1999–2001, which is the exact period in which the market earns its “casino” reputation.
- Beginning around 2003–2004, China's predicted variation of future profit from prices rises sharply.
- For many post-2004 years, China's series is close to, and sometimes above, the US series for both $k = 3$ and $k = 5$.
- The dotted rejection-region boundary shows that in most post-2004 years one cannot reject the hypothesis that China is as informative as the US.

Economic message. The main contribution of the paper is already visible here: China's stock market stops looking like a purely noisy gambling venue and starts looking like an information-aggregating market.

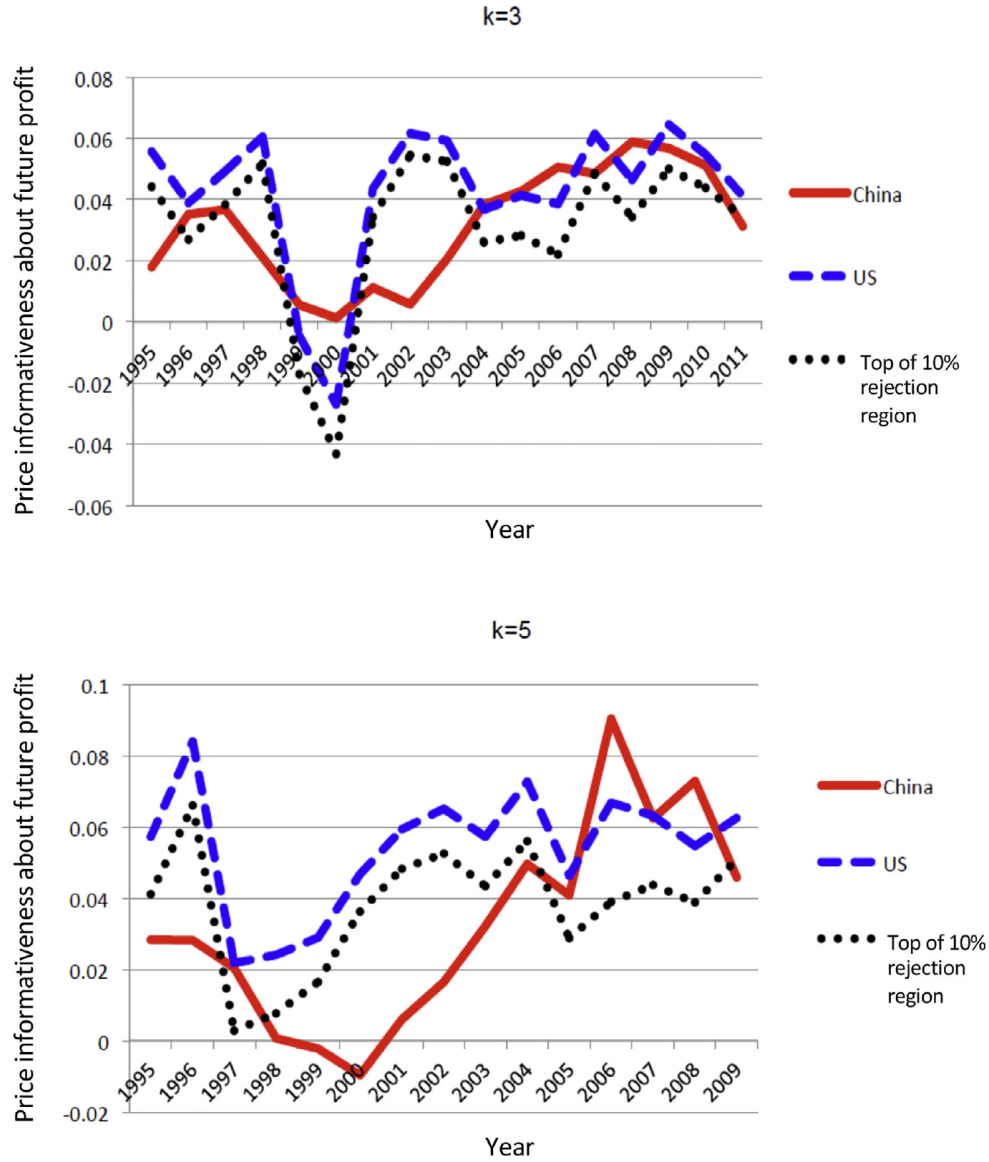


Fig. 2. Stock price informativeness about future profit: China versus US. The solid and dashed lines plot the predicted variation $b_t \times \sigma_t(\log(\frac{M_{i,t}}{A_{i,t}}))$ from annual regressions of the form
$$\frac{E_{i,t+k}}{A_{i,t}} = a_t + b_t \log\left(\frac{M_{i,t}}{A_{i,t}}\right) + c_t \left(\frac{E_{i,t}}{A_{i,t}}\right) + d_t^s \mathbb{1}_{i,t}^s + \varepsilon_{i,t+k}$$
 for China and the US. The dotted line shows the highest China price informativeness level for which the hypothesis that prices in China are as informative as in the US can be rejected at the 10% level in a one-sided test.

Figure 1: Figure 2 from the paper: stock price informativeness about future profit, China versus US.

5.2 Figure 4: the rise in informativeness lines up with reform

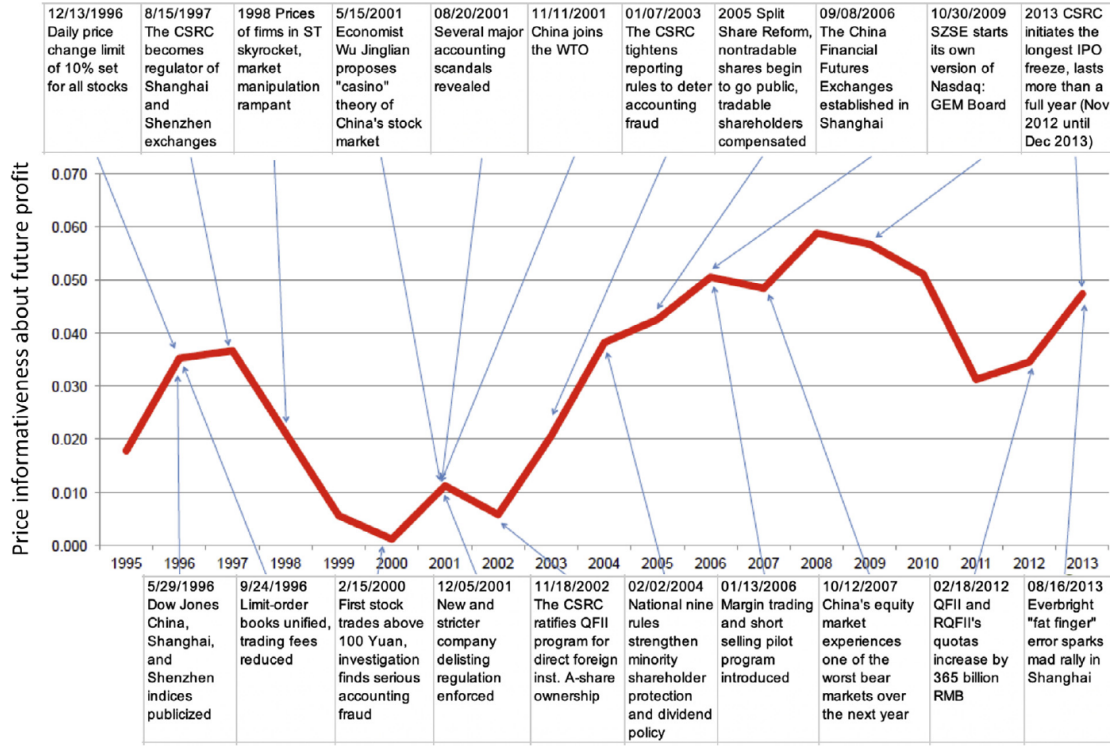
Read:

- Informativeness is low during the late-1990s period associated with rampant manipulation, accounting concerns, and the “casino” label.
- The measure rises after WTO accession, stricter delisting enforcement, the National Nine Rules,

and especially the 2005 Split-Share Structure Reform.

- Informativeness weakens somewhat after the global crisis, but remains far above the trough years.

Economic message. The paper is not claiming that one reform alone “caused” better prices. Rather, the figure shows that informativeness improved during a broad modernization of the market’s investor base, disclosure environment, and trading technology.



g. 4. Stock price informativeness, regulatory reforms, and news events.
redicted variation $b_t \times \sigma_t(\log(\frac{M_{i,t}}{A_{i,t}}))$ from annual cross-sectional regressions of the form

$$\frac{i_{i,t+k}}{A_{i,t}} = a_t + b_t \log\left(\frac{M_{i,t}}{A_{i,t}}\right) + c_t \left(\frac{E_{i,t}}{A_{i,t}}\right) + d_t^s 1_{i,t}^s + \varepsilon_{i,t+k}$$

or forecasting horizon $k = 3$ and the timing of various reforms and events that plausibly affected this predicted variation.

Figure 2: Figure 4 from the paper: stock price informativeness, regulatory reforms, and news events.

5.3 Table 2: state ownership becomes a major drag after 2008

Read:

- Before 2009, the interaction between state ownership and price informativeness is small: $b_1 = 0.009$ for $k = 3$ and 0.012 for $k = 5$.
- After 2008, the interaction turns sharply negative: $b_2 = -0.055$ for $k = 3$ and -0.115 for $k = 5$.
- The subperiod change is economically large: $b_2 - b_1 = -0.064$ for $k = 3$ and -0.127 for $k = 5$.
- The year-by-year cross-sectional averages tell the same story, so the result is not just coming from pooled time-series averaging.

Economic message. The problem is not “China” in general. The problem is that after the crisis, firms with high state ownership become much harder to value using prices because subsidies and policy goals weaken the mapping from prices to future profits.

J.N. Carpenter, F. Lu and R.F. Whitelaw / Journal of Financial Economics 139 (2021) 679–696

Table 2

State ownership and stock price informativeness about future profit.

The table shows estimates of the coefficients b_1 , b_2 , and $b_2 - b_1$ along with their White-heteroscedasticity-consistent t -statistics (in parentheses) from panel regressions of the form

$$\frac{E_{i,t+k}}{A_{i,t}} = a_{0t} + a_{1t}X_{i,t} + (b_{0t} + (b_1 1_{t < 2009} + b_2 1_{t \geq 2009})X_{i,t}) \log\left(\frac{M_{i,t}}{A_{i,t}}\right) + c_t\left(\frac{E_{i,t}}{A_{i,t}}\right) + d_t^s 1_{i,t}^s + \varepsilon_{i,t+k},$$

where $X_{i,t}$ is the firm's fraction of shares that are state-owned, for China for forecasting horizons $k = 3$ and 5 over the period 1995 to 2016 – k . The row labeled “Cross-sectional” contains the corresponding subperiod averages of estimates of yearly interaction coefficients b_{1t} from annual cross-sectional regressions of the form

$$\frac{E_{i,t+k}}{A_{i,t}} = a_{0t} + a_{1t}X_{i,t} + (b_{0t} + b_{1t}X_{i,t}) \log\left(\frac{M_{i,t}}{A_{i,t}}\right) + c_t\left(\frac{E_{i,t}}{A_{i,t}}\right) + d_t^s 1_{i,t}^s + \varepsilon_{i,t+k}.$$

	$k = 3$			$k = 5$		
	b_1	b_2	$b_2 - b_1$	b_1	b_2	$b_2 - b_1$
	0.009 (1.63)	−0.055 (−6.79)	−0.064 (−6.52)	0.012 (1.29)	−0.115 (−7.18)	−0.127 (−7.07)
Cross-sectional	0.012	−0.062	−0.075	−0.004	−0.132	−0.128

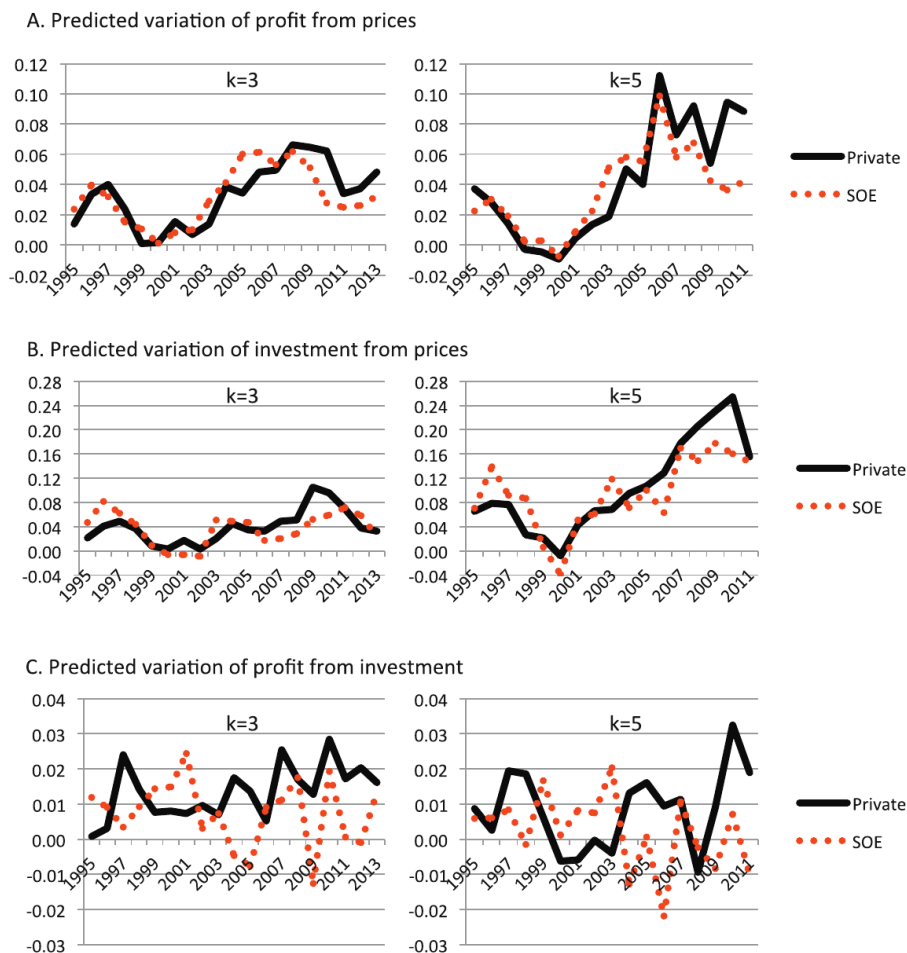
Figure 3: Table 2 from the paper: state ownership and stock price informativeness about future profit.

5.4 Figure 5: private firms use the market more than SOEs

Read:

- Panel A shows the private-SOE gap in price informativeness opening after 2008.
- Panel B shows that for private firms, prices increasingly forecast future investment; for SOEs the relationship is weaker.
- Panel C shows that investment by private firms becomes more informative about future profit, while the same measure is flat or weaker for SOEs.
- The paper quantifies these co-movements: the time-series correlation between average price informativeness and investment-price sensitivity is 84% for private firms versus 52% for SOEs, and the correlation between price informativeness and allocational efficiency is 51% for private firms versus -37% for SOEs.

Economic message. This is the strongest evidence in the paper that prices matter for real decisions. The market looks informative for private firms precisely in the sense that managers appear to act on those signals. That link largely breaks for SOEs after the stimulus period.



5. Privately owned firms versus SOEs in China. Privately owned firms are defined as those with less than or equal to 40% of equity owned by the state, and SOEs are defined as those with more than 40% state ownership. Panels A, B, and C show predicted variations from annual regressions specified in Eqs. (1), (2), and (3), respectively, for each subsample of

Figure 4: Figure 5 from the paper: privately owned firms versus SOEs in China.

5.5 Tables 8 and 9: China's market offers high returns and low correlation, which implies a high cost of capital

Read:

- Table 8 reports an annualized excess return of 12.76% in CNY for China, versus 7.83% for the US. In USD terms China's excess return is 14.77%, again well above the US.
- China's market volatility is about double that of the US, but its correlation with the US market is only 0.19, and correlations with Europe and Japan are similarly low.
- Table 9 shows that China's market earns monthly alphas of about 0.8%–1.0% relative to US and global factor models.

Economic message. From a corporate-finance perspective, these are the fingerprints of under-integration. Chinese firms appear to face a higher required return because domestic investors are poorly diversified and foreign investors still face enough barriers that they underweight China.

Table 8

Excess returns on stock markets in large economies 1995–2016.

The table shows annualized means and volatilities (in %) of monthly excess returns in stock markets in four large economies and their correlations over the period January 1995 to December 2016.

	China CNY	China USD	US USD	Europe USD	Japan USD
Mean	12.76	14.77	7.83	6.44	0.24
Volatility	31.53	31.63	15.32	17.51	17.95
Corr. with US		0.19			
Corr. with Europe		0.23	0.80		
Corr. with Japan		0.13	0.45	0.50	

(a) Table 8: market excess returns and correlations.**Table 9**

Alphas of China's stock market with respect to US and global factors 1995–2016.

The table shows monthly alphas (in %) of USD returns on China's tradable-value-weighted stock market portfolio with respect to the US and global Fama–French factors, and their Newey–West adjusted t -statistics (in parentheses) over the period January 1995 to December 2016.

US factors			Global factors		
1-factor	3-factor	5-factor	1-factor	3-factor	5-factor
0.97	0.97	0.90	0.99	0.90	0.81
(1.39)	(1.35)	(1.25)	(1.47)	(1.28)	(1.14)

(b) Table 9: China's alphas relative to US and global factors.**Figure 5:** Cost-of-capital evidence from the paper.

6 Short summary

- The paper asks whether China's stock market is informative enough to play a real capital-allocation role.
- Using Bai et al. (2016)-style predicted-variation measures, it shows that since about 2004 Chinese stock prices have been roughly as informative about future profits as US stock prices.
- For private firms, more informative prices go together with stronger price-investment sensitivity and better allocational efficiency, consistent with managers learning from prices.
- For high-state-ownership firms, especially after the 2008–2009 stimulus, both price informativeness and investment efficiency deteriorate.
- Dual listing in Hong Kong is associated with lower A-share informativeness, while QFII ownership is weakly positively associated with informativeness.
- Realized return evidence suggests Chinese firms may face a higher cost of equity capital than US firms because of domestic under-diversification and incomplete global integration.

7 Conclusion

7.1 Core conclusion

The paper's central conclusion is that China's stock market has more real economic value than its reputation suggests. It does not merely generate volatile prices; for a large and important subset of firms, it aggregates information about future profitability and appears to guide investment in a more efficient direction.

7.2 Economic intuition

A market can matter for capital allocation in two distinct ways. First, prices can forecast future cash flows. Second, managers can learn from those prices and alter investment accordingly. The paper

finds both channels for private firms in China. For SOEs, however, subsidies, political objectives, and state-directed credit weaken both the predictability of profits and the willingness or ability of managers to follow market signals.

7.3 Takeaway

The most important implication is not that China's stock market is already fully efficient. It is that the market's main bottleneck is institutional. Further broadening the investor base, reducing barriers to foreign participation, and relying less on SOEs as policy instruments would likely increase the market's contribution to capital allocation and lower firms' cost of capital.